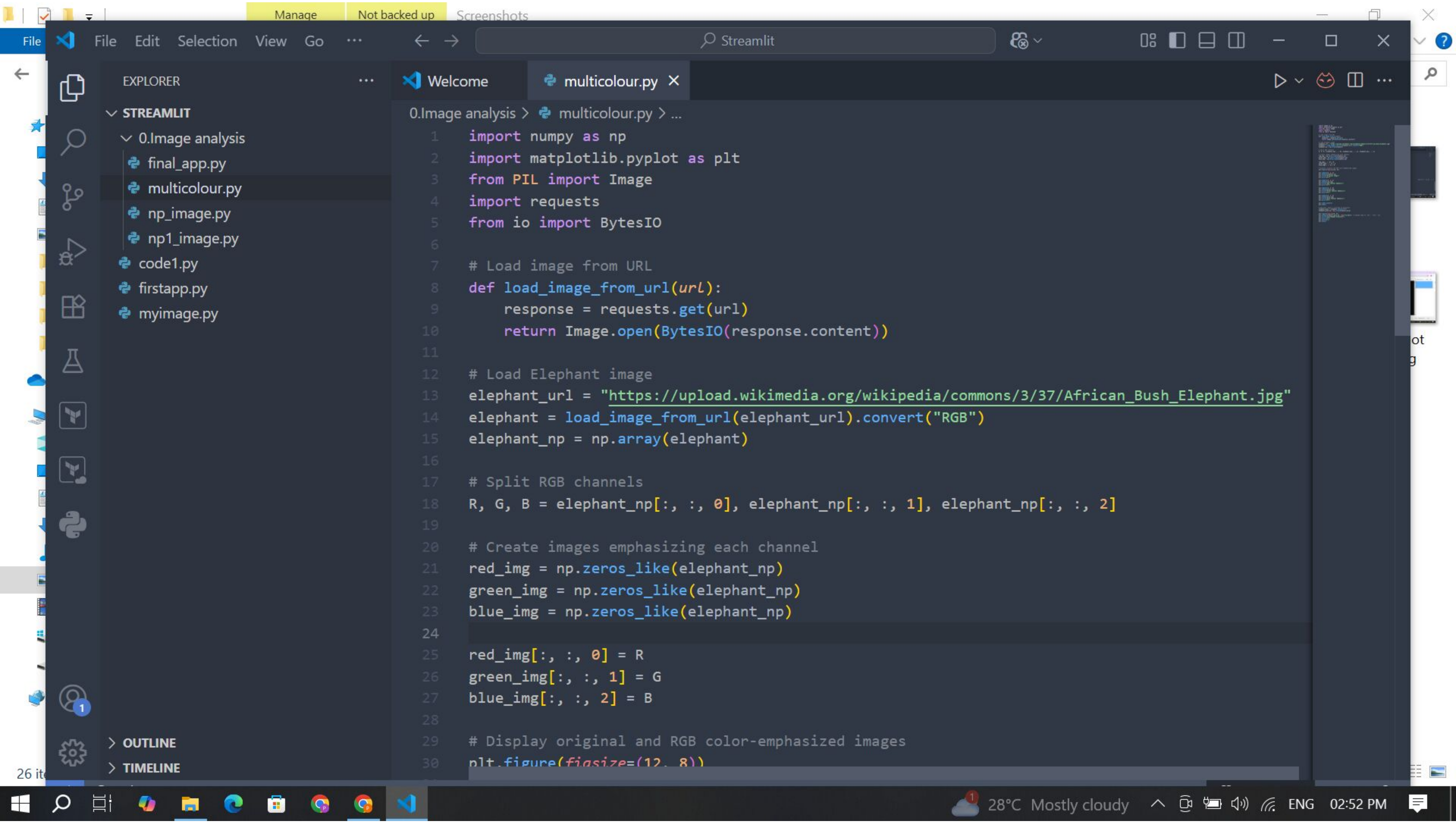

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Streamlit
EXPLORER
STREAMLIT
0.Image analysis
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np_image.py
np1_image.py
code1.py
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OUTLINE
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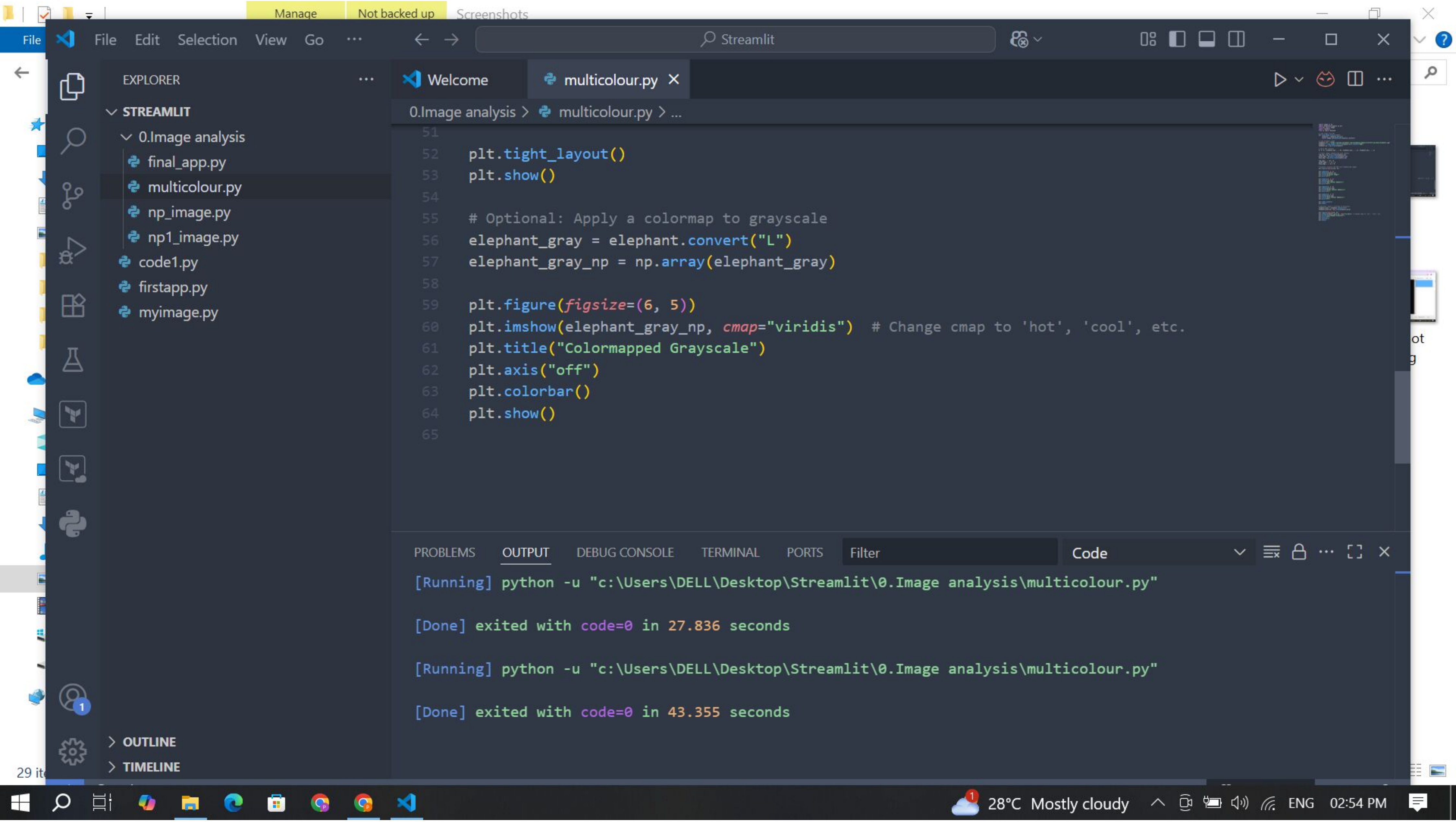
Welcome
multicolour.py
0.Image analysis > multicolour.py > ...
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image
import requests
from io import BytesIO
# Load image from URL
def load_image_from_url(url):
response = requests.get(url)
return Image.open(BytesIO(response.content))
# Load Elephant image
elephant_url = "https://upload.wikimedia.org/wikipedia/commons/3/37/African_Bush_Elephant.jpg"
elephant = load_image_from_url(elephant_url).convert("RGB")
elephant_np = np.array(elephant)
# Split RGB channels
R, G, B = elephant_np[:, :, 0], elephant_np[:, :, 1], elephant_np[:, :, 2]
# Create images emphasizing each channel
red_img = np.zeros_like(elephant_np)
green_img = np.zeros_like(elephant_np)
blue_img = np.zeros_like(elephant_np)
red_img[:, :, 0] = R
green_img[:, :, 1] = G
blue_img[:, :, 2] = B
Display original and RGB color-emphasized images
28°C Mostly cloudy
ENG
02:52 PM

```python
# Display original and RGB color-emphasized images
plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(elephant_np)
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(red_img)
plt.title("Red Channel Emphasis")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(green_img)
plt.title("Green Channel Emphasis")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(blue_img)
plt.title("Blue Channel Emphasis")
plt.axis("off")

plt.tight_layout()
plt.show()

# Optional: Apply a colormap to grayscale
elephant_gray = elephant.convert("L")
elephant_gray_np = np.array(elephant_gray)
```



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0.Image analysis > multicolour.py > ...
plt.tight_layout()
plt.show()
# Optional: Apply a colormap to grayscale
elephant_gray = elephant.convert("L")
elephant_gray_np = np.array(elephant_gray)
plt.figure(figsize=(6, 5))
plt.imshow(elephant_gray_np, cmap="viridis") # Change cmap to 'hot', 'cool', etc.
plt.title("Colormapped Grayscale")
plt.axis("off")
plt.colorbar()
plt.show()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Filter
Code
[Running] python -u "c:\Users\DELL\Desktop\Streamlit\0.Image analysis\multicolour.py"
[Done] exited with code=0 in 27.836 seconds
[Running] python -u "c:\Users\DELL\Desktop\Streamlit\0.Image analysis\multicolour.py"
[Done] exited with code=0 in 43.355 seconds
28°C Mostly cloudy
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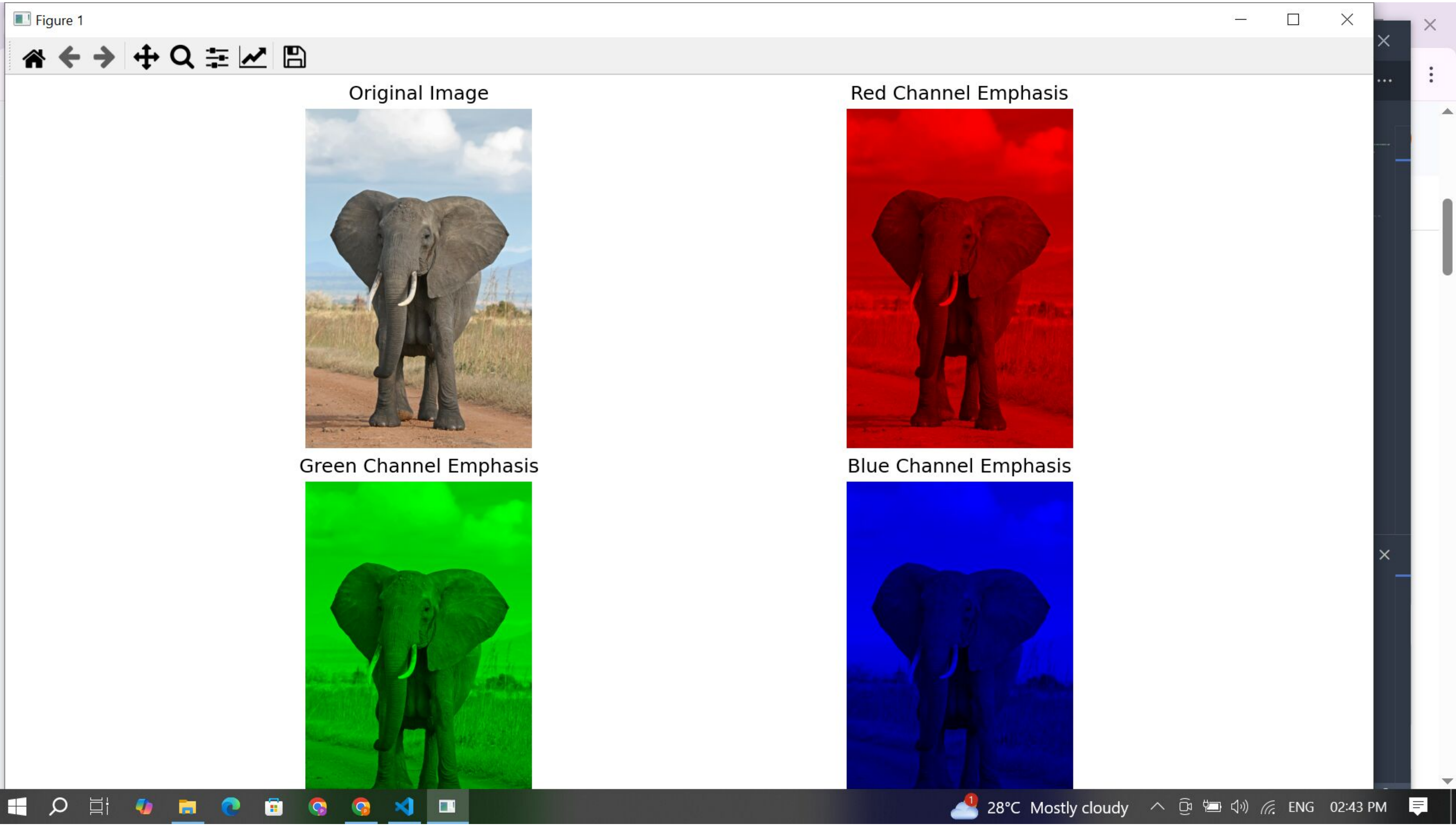
Figure 1
Original Image
Red Channel Emphasis
Green Channel Emphasis
Blue Channel Emphasis
28°C Mostly cloudy
ENG
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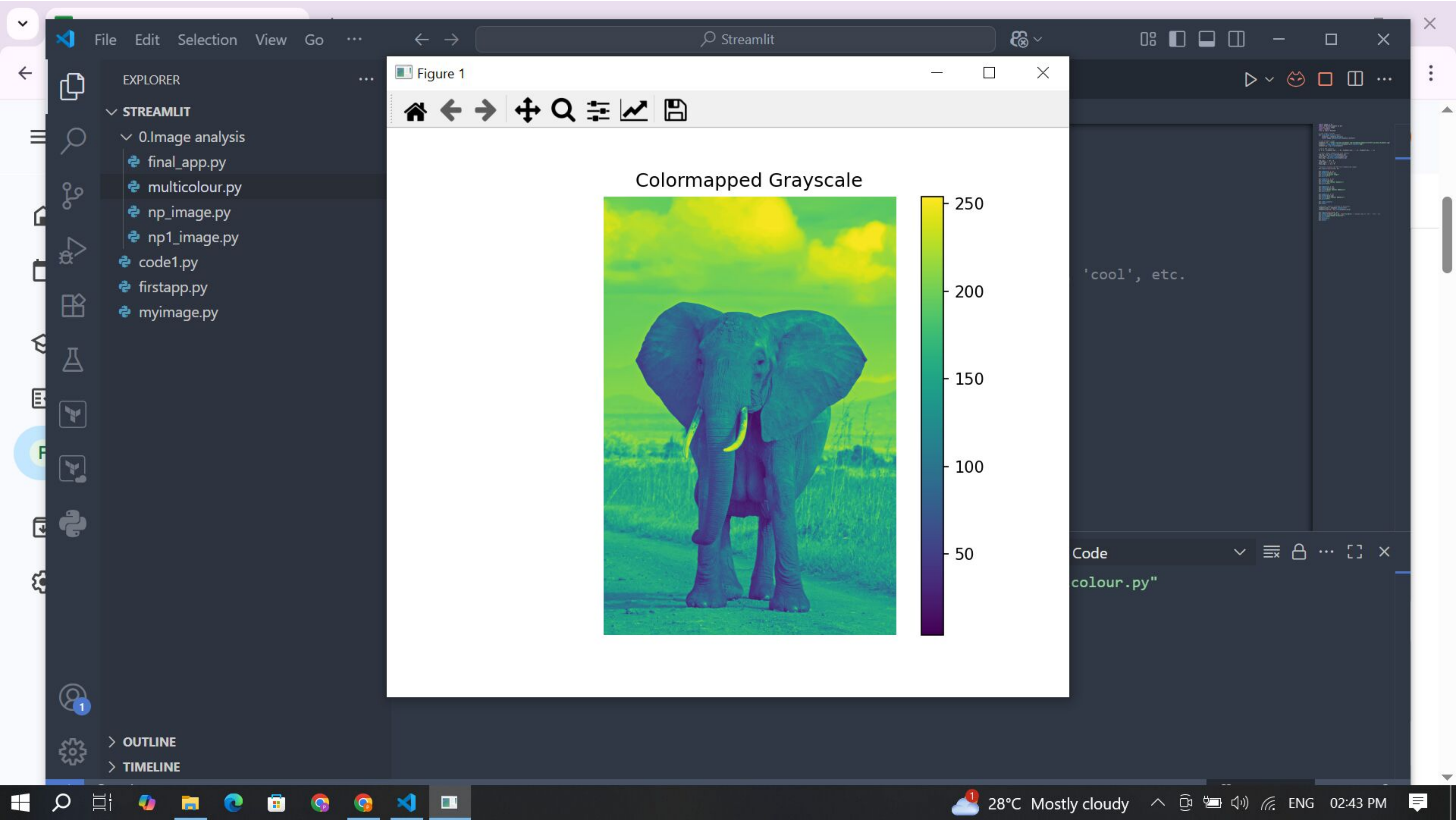
File Edit Selection View Go ...
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OUTLINE
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Figure 1
Colormapped Grayscale
250
200
150
100
50
'cool', etc.
Code
colour.py"
28°C Mostly cloudy
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